

Complexity Analysis – Government Database BST

Operation	Best Case	Average Case	Worst Case
BST Search	$O(1)$	$O(\log n)$	$O(n)$
Linear Search	$O(1)$	$O(n)$	$O(n)$
BST Insertion	$O(1)$	$O(\log n)$	$O(n)$

Space Complexity: The BST requires $O(n)$ space for n nodes.

Observed results: BST comparisons for A7, B3 and A120 were 3, 6 and 3. Linear search comparisons were 3, 7 and 6. The original insertion order produced height 6; sorted and reverse insertion produced height 8.

Interpretation: A balanced BST approaches $O(\log n)$ search, while an unbalanced BST can degrade to $O(n)$. A self-balancing AVL Tree or Red-Black Tree is therefore suitable for a growing database.